where

$$\Delta M^0 = M_3^0 + M_4^0 - M_1^0 - M_2^0 \tag{10.4}$$

The reaction energy may thus be computed from the rest masses of the reactants and products. If mass disappears in the reaction ($\Delta M^0 < 0$), energy is released: the reaction is said to be *exoergic*, and Q is positive. For $Q < 0$ the reaction is *endoergic* and $\Delta M^0 > 0$. (For comparison, in chemistry a negative value of ΔH is associated with exothermic reactions.) Tables of Q-values, especially for light projectiles, can be found in several literature sources.

The relativistic mass equation (5.26)

$$E_{\text{kin}} = \left(m - m^0\right)\mathbf{c}^2 \tag{10.5}$$

can be separated into two terms if we define

$$E_{\text{mass}}^0 \equiv m^0\mathbf{c}^2 \tag{10.6}$$

and

$$E_{\text{tot}} \equiv m\mathbf{c}^2 \tag{10.7}$$

Then

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{mass}}^0 \tag{10.8}$$

We will call E_{mass}^0 the *mass energy*, which is independent of the kinetic energy of the particle. E_{mass}^0 is *potential* energy, and can in principle be converted into any other energy form because (10.6) is a form of Einstein's mass-energy relation. E_{mass}^0 is closely related to the nuclear binding energy (cf. §4.4).

In the previous section we pointed out that for $Q > 0$ the reaction is exoergic and, as a consequence of that, mass has to disappear ($\Delta M^0 < 0$). If the total energy of the system is constant (conservation rule (a)), it becomes obvious from (10.8) that when E_{mass}^0 decreases E_{kin} must increase. Thus the products of the exoergic nuclear reaction have a higher kinetic energy than the reactants.

10.3. THE COULOMB BARRIER

Equation (10.8) is a special case of a more general equation applicable to all systems of particles:

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{pot}} \tag{10.9}$$

where E_{kin} = translational, rotational, vibrational, etc., energy and E_{pot} = mass energy, gravitational, electrostatic energy, surface energy, chemical binding energy, etc. This is the total energy referred to in (5.2).

In nuclear reactions we would include the mass energy for the atomic masses in their ground state E^0_{mass}, the excitation energy of the nucleus above its ground state E_{exc}, the absorption or emission of photons in the reaction E_υ, and − in reactions between charged particles − the electrostatic potential (Coulomb) energy E_{coul}. Since we are concerned only with reactions induced by neutral or positively charged particles, the Coulomb energy is zero or positive (i.e. repulsive). If charged, the incoming projectile must possess sufficient kinetic energy to overcome possible repulsion. During the intermediate stage, this energy appears as partly as recoil energy of the center of mass and partly as potential energy. In the second step of the reaction the repulsion of charged products results in greater kinetic energy. Thus, generally,

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{coul}} + E^0_{\text{mass}} + E_{\text{exc}} + E_\upsilon \tag{10.10}$$

In nuclear reactions the total energy must be conserved, although the distribution of this energy in the different forms of (10.10) usually changes during the course of the reaction. For example, any decrease in the mass energy term must be balanced by a complimentary increase in one or more of the other energy terms. An example of this occurs in the explosion of nuclear weapons where a fraction of the rest mass is transformed to other forms of energy. The opposite reaction, the transformation of kinetic energy to mass, occurs in the production of elementary particles and in high energy acceleration of particles in cyclotrons, synchrotrons, etc.

Let us, as an example of the use of (10.10), consider a reaction of a positively charged particle (M_1, Z_1, v_1) with a target atom (M_2, Z_2, $v_2 \approx 0$). We make two simplifying assumptions: the target nucleus is in the center of the coordinate system (i.e. $v_2 = 0$) and relativistic mass corrections can be neglected.

Because both projectile and target have positive charge they must repel each other according to the Coulomb law:

$$F_{\text{coul}} = k\,\mathbf{e}\,Z_1\,\mathbf{e}\,Z_2/x^2 \tag{10.11}$$

where k is 8.99×10^9 (N m^2 C^{-2}). This force is shown as a function of the distance between the particles in Figure 10.2a. At a distance greater than r_{n} only the Coulomb repulsive force, F_{coul}, is in operation; however, for distances less than r_{n} both the attractive nuclear force F_{nucl} and the repulsive Coulomb force F_{coul} act upon the system. The total force is given by $F_{\text{tot}} = F_{\text{coul}} + F_{\text{nucl}}$. This is shown by a solid line in the Figure. At some particular distance designated r_{c}, the forces balance each other and at shorter distances ($x < r_{\text{c}}$), the attractive nuclear force dominates. The distance r_{c} is known as the Coulomb radius; to be more exact, r_{c} is often considered to be the sum of the projectile and target radii.

In Figure 10.2b the variation in the value of the different forms of (10.10) is indicated as a function of the distance x between the two particles. At long distances from the target

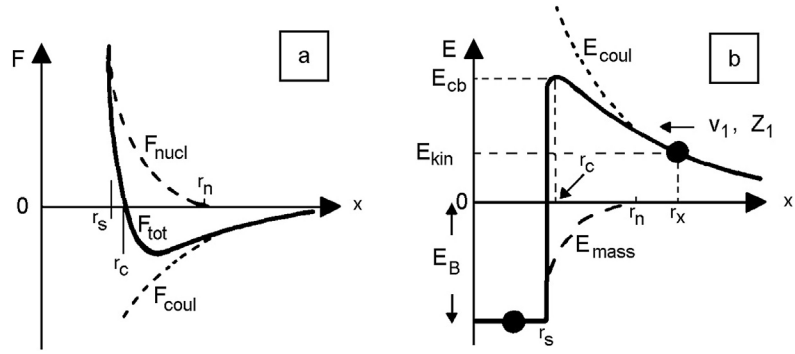


Figure 10.2 Forces (a) and energy (b) conditions when a charged projectile (Z_1, v_1) reacts with a target nucleus. In (a) the coulomb repulsion force is directed away from the nucleus, hence it is shown as negative, and the nuclear force is directed towards the nucleus and shown as positive.

nucleus the kinetic energy of the projectile is decreased due to the Coulomb repulsion. For such distances the nuclear force can be neglected and

$$E^0_{\text{kin}} = E_{\text{kin}} + E_{\text{coul}} \tag{10.12}$$

Where E^0_{kin} is the original kinetic energy of the projectile and E_{coul} is the electrostatic (Coulomb) potential energy. The initial projectile energy is thus divided into remaining kinetic enegy, E_{kin} and potential coulomb enegy, E_{coul}, which at the distance from the center x can be written as

$$E_{\text{coul}} = k\, Z_1\, Z_2\, \mathbf{e}^2/x \tag{10.13}$$

As the projectile approaches the target nucleus, the Coulomb repulsion causes the potential energy to increase as the kinetic energy of the particle decreases. If this decrease in kinetic energy of the particle is such that the kinetic energy reaches a value of zero at any distance greater than r_c, the particle is reflected away from the nucleus before it is close enough to experience the attractive nuclear force. The projectile is thus hindered by a Coulomb potential barrier from causing nuclear reaction. A necessary condition for charged projectiles to cause nuclear reactions is that E_{in} exceeds the Coulomb barrier height, E_{cb}, defined as:

$$E_{\text{cb}} = k\, Z_1\, Z_2\, \mathbf{e}^2/r_c \tag{10.14}$$

where r_c is the *Coulomb radius*. This equation is useful for determining the Coulomb radius ("distance of closest approach") of nuclei; for conservation of momentum see (10.15 and 10.16).

We learned in Chapter 4 that experimentally, the nuclear radius is given by $r = r_o\, A^{1/3} \times 10^{-15}$ m. Thus far we have discussed three different kinds of nuclear radius constants – r_s, r_c, and r_n. In addition we have treated the target and the projectile as points in space.

Experiments have indicated that the values of r_o are approximately 1.1 for the radius r_s of constant nuclear density, 1.3 for the Coulomb radius r_c, and 1.4 for the nuclear radius r_n, which includes surface effects.

To calculate the value of the energy of the Coulomb barrier we can use a model in which the target nuclei and the projectile are just touching so that r_c is taken as the sum of the radii of the projectile and the target nucleus. Moreover, we must now consider that the center of mass is not immobile in the collision but has a certain kinetic energy determined by the conservation of momentum. With this model the Coulomb barrier energy is given by the equation

$$E_{cb(min)} = 1.109\,(A_1 + A_2)\left[Z_1 Z_2 / \left\{A_2\left({A_1}^{1/3} + {A_2}^{1/3}\right)\right\}\right]\ (\text{MeV}) \qquad (10.15)$$

$E_{cb(min)}$ is the minimum energy that a projectile of mass A_1 and charge Z_1 must have in order to overcome the Coulomb barrier of a target nucleus of mass A_2 and charge Z_2 in a central collision. For the reaction between an α-particle and the nucleus ^{14}N, $E_{cb(min)}$ has a value of 4.99 MeV; it is obvious that Rutherford's α-particles of 7.68 MeV from the decay of ^{210}Po had sufficient kinetic energy to cause reaction.

In Figure 10.3 the values of $E_{cb(min)}$ for protons, α-, and ^{16}O-particles are shown as a function of the atomic number of the target nucleus. The A-values of the target are of the most stable isotope of the element Z. The initial decrease in the curves occur because in that range the projectile is heavier than the target nuclei.

In most cases the projectile-target collision is not head-on. Consider a projectile that would pass the target nucleus with a shortest distance, x, between their centers if no forces acted between them. This distance is called the *impact parameter*. The repulsive Coulomb force will cause the projectile to pass at a larger distance. By considering conservation of momentum and energy, and assuming that the distance of closest approach, d, is outside the range of the nuclear force one obtains

$$d = r + \left[r^2 + x^2\right]^{1/2} \qquad (10.16)$$

In this equation we have introduced the *collision radius*, r, defined as

$$r = \mathrm{k}\ Z_1\ Z_2\ \mathbf{e}^2 / \left[m_{\mathrm{red}}\left(v_1^0\right)^2\right] \qquad (10.17)$$

where $\mathrm{k} = 8.99 \times 10^9$ N m^2 C^{-2}, $m_{red}^{-1} = m_1^{-1} + m_2^{-1}$ and v_1^0 is the initial projectile velocity in the laboratory coordinate system.

A technique called *Coulomb excitation* is used to induce rotationally excited states in nuclei; an example of this was shown in Figure 6.7. In order to impart high energy to the nucleus without causing nuclear transformation, heavy ions with kinetic energy below that required for passing over the Coulomb barrier ($E_1 < E_{cb(min)}$) may be used as projectiles.

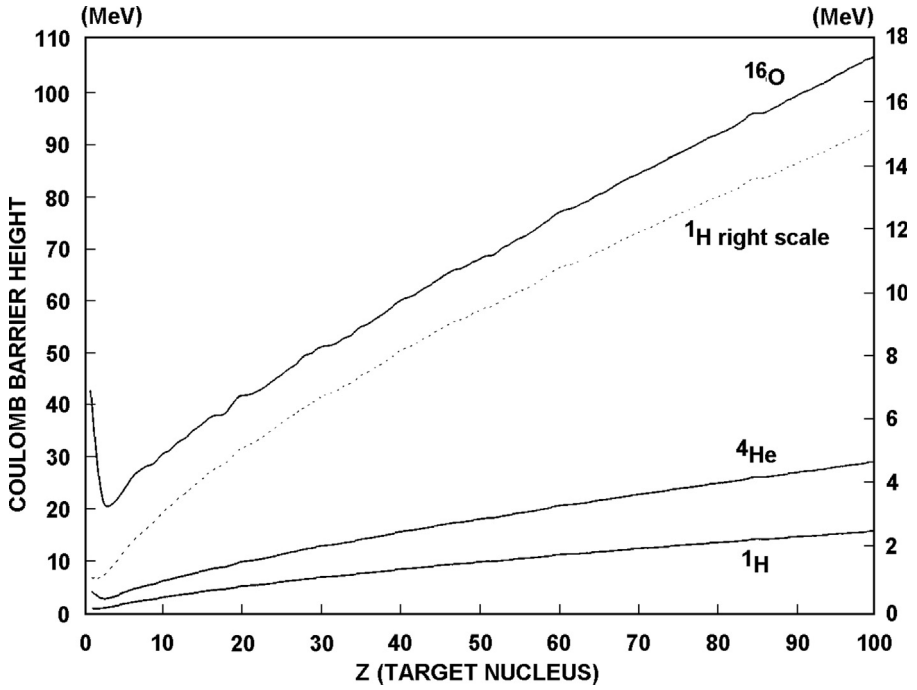


Figure 10.3 The Coulomb barrier height, E_{cb}(min) according to (10.15) for reactions between a target element Z (of most common A) and projectiles ^{1}H, ^{4}He, and ^{16}O.

10.4. RUTHERFORD SCATTERING

If a collimated beam of particles (Z_1, A_1, $E_{\text{kin},1}$) strikes a foil (Z_2, A_2) so that most of the particles pass through the foil without any reduction in energy, it is found that many particles are scattered away from their incident direction (Fig. 10.4). We can neglect multiple scattering and nuclear transformations, since they are many times less than the number of scattering events. Solid angles are measured in steradians, where 1 steradian corresponds to $1/(4\pi)$:th of the surface area of an enclosing sphere such that all particles will cross its surface area.

Geiger and Marsden found that scattering follows the relation

$$\mathrm{d}\sigma/\mathrm{d}\Omega = \left[k\ Z_1\ Z_2\ \mathbf{e}^2/\left\{2m_{\text{red}}\left(v_1^0\right)^2\right\}\right]^2/\sin^4(\theta/2) \tag{10.18}$$

Where $\mathrm{d}\sigma/\mathrm{d}\Omega$ is the differential cross-section (reaction cross-section in barns or m^2 per steradian), θ is the scattering angle (Fig. 10.4) and v_1^0 incident particle velocity. The differential cross-section is a measure of the probability σ of a scattering event per unit solid angle Ω.

$$\mathrm{d}\sigma/\mathrm{d}\Omega = n/(n_{\text{o}}\ N_{\text{v}}\ x\ \Delta\Omega) \tag{10.19}$$